Richard Wu

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Little Neck, NY 11362

BACKEND SOFTWARE ENGINEER

PROFESSIONAL EXPERIENCE

Amazon Web Services **2020 - 2023**

Software Engineer

- Improved security posture and scaled Session Manager, a highly available, real-time, distributed compute management solution reliant on persistent websocket connections
- Contributed to a team migration from monolith to control-plane/data-plane architecture; owned control-plane provisioning workflows and rollout automation
- Built data synchronization pipeline that reduced manual intervention of session desync by 30%
- Helped implement testing strategy expansion that reduced software regressions and increased developer velocity
- Expanded operational health insights, logging and dashboards in order to decrease developer time spent on triage and investigation and increase observability
- Participated in incident response and postmortems; shipped follow-up fixes to prevent recurrence
- Improved runbooks and operational tooling to reduce toil during incidents

PROJECT WORK

Executive Order

https://www.kaggle.com/code/penglingwu/executive-order

White House Executive Order Digestion and Retrieval

- Chat with a LLM about recent political policy
- Designed retrieval, embeddings, and search pipeline for documents to be used for conversation context
- Tech stack: Python, Langchain, FAISS for similarity search

littleblackbook.io

https://github.com/Rich-Wu/datetracker

Date tracking and social app

- Full-stack web app for logging interactions, profile management, and trend analytics
- Analytics dashboard for behavioral metrics and time-series insights
- Deployed on Google App Engine; REST API in Go/Gin; MongoDB persistence
- Tech stack: Go, Gin, MongoDB, Tailwind, Webpack, Google AppEngine

Mandarin Learner

https://github.com/Rich-Wu/wbxh

Mandarin Chinese translator and learning tool

- Webapp that takes in an image of Chinese text, recognizes the characters, translates and breaks down the text to aid in learning the language
- Tech stack: OpenAI API, Typescript, TesseractJS, HTML, CSS, Vite

SKILLS

Java	MongoDB	ReactJS
Go	AWS Cloud	Docker
Websockets	Python	Rest APIs

EDUCATION

B.S. Computer Science

University of California, Riverside